

LOGISTIC REGRESSION

A study of customer churn rate using their consumption behavior as predictors in
E-commerce

AD571 SPR 2026

Team 1: Antara Sudhir, Hye Won Kim, Julie Tran

TABLE OF CONTENT

1	Introduction - Database
2	ETL
3	What is Logistic Regression
4	Data Feature Enginneering
5	Model Results
6	Interpretation
7	Conclusion

Dataset Overview

Multi-channel E-commerce with Behavioral Metrics

CustomerID	Customer_Segment	Acquisition_Channel	Age	Gender	Job_Category	Annual_Income	Education_Level	Marital_Status	Household_Size	Churned
C00001	Regular	2023-12-05	49	Female	Manufacturing	53900	Bachelor's	Married	2	0
C00002	New	2023-01-26	39	Male	IT	47557	High School	Single	1	0
C00003	Regular	2023-07-01	51	Female	Other	81597	High School	Widow	2	0
C00004	Regular	2023-09-22	64	Male	Finance	100299	Bachelor's	Married	3	0
C00005	New	2023-09-15	38	Female	Other	132438	Master	Married	2	0
C00006	VIP	2023-08-08	38	Male	Manufacturing	73299	High School	Single	1	0
C00007	New	2023-10-01	65	Non-binary	Finance	101885	Master	Married	1	0
C00008	VIP	2023-03-05	53	Male	Education	107531	Master	Married	3	0
C00009	New	2023-01-30	34	Female	Retail	49696	Bachelor's	Single	1	0
C00010	New	2023-03-24	50	Male	Education	47504	Bachelor's	Married	2	0
C00011	Regular	2023-06-19	35	Female	Manufacturing	30856	Master	Divorced	2	0
C00012	New	2023-04-19	35	Female	Healthcare	25000	High School	Single	1	0
C00013	Regular	2023-02-18	45	Female	Finance	25000	Bachelor's	Single	2	0
C00014	Regular	2023-03-11	18	Female	IT	25000	PhD	Married	1	0
C00015	Regular	2023-10-29	18	Female	Other	67102	High School	Married	3	0
C00016	New	2023-04-01	33	Female	Healthcare	25000	High School	Single	1	0
C00017	Regular	2023-08-05	26	Male	Healthcare	100983	PhD	Married	1	0
C00018	Regular	2023-08-03	46	Non-binary	Education	41067	High School	Married	2	0
C00019	Regular	2023-03-07	28	Male	Education	44293	PhD	Married	3	0
C00020	Regular	2023-06-11	20	Male	Other	46670	Master	Married	4	0
C00021	VIP	2023-05-20	63	Female	Other	77927	High School	Single	1	0
C00022	VIP	2023-09-11	38	Male	Healthcare	88644	High School	Married	1	0
C00023	New	2023-00-10	42	Female	Detail	25000	PhD	Married	1	0

Unit of analysis

Each row represents a customer record with behavioral, transactional, and engagement-related information

What dataset includes:

- Customer demographic information
- Purchase and payment behavior
- Loyalty and engagement metrics
- Retention-related variables, including a binary churn indicator

Why we selected this dataset

- The binary Churned variable makes the dataset suitable for logistic regression
- Behavioral, purchase, loyalty, and engagement variables can help explain churn risk
- The analysis can support customer retention and targeted re-engagement strategies

VARIABLE SELECTION

VARIABLE SELECTION FOR **CHURN ANALYSIS**

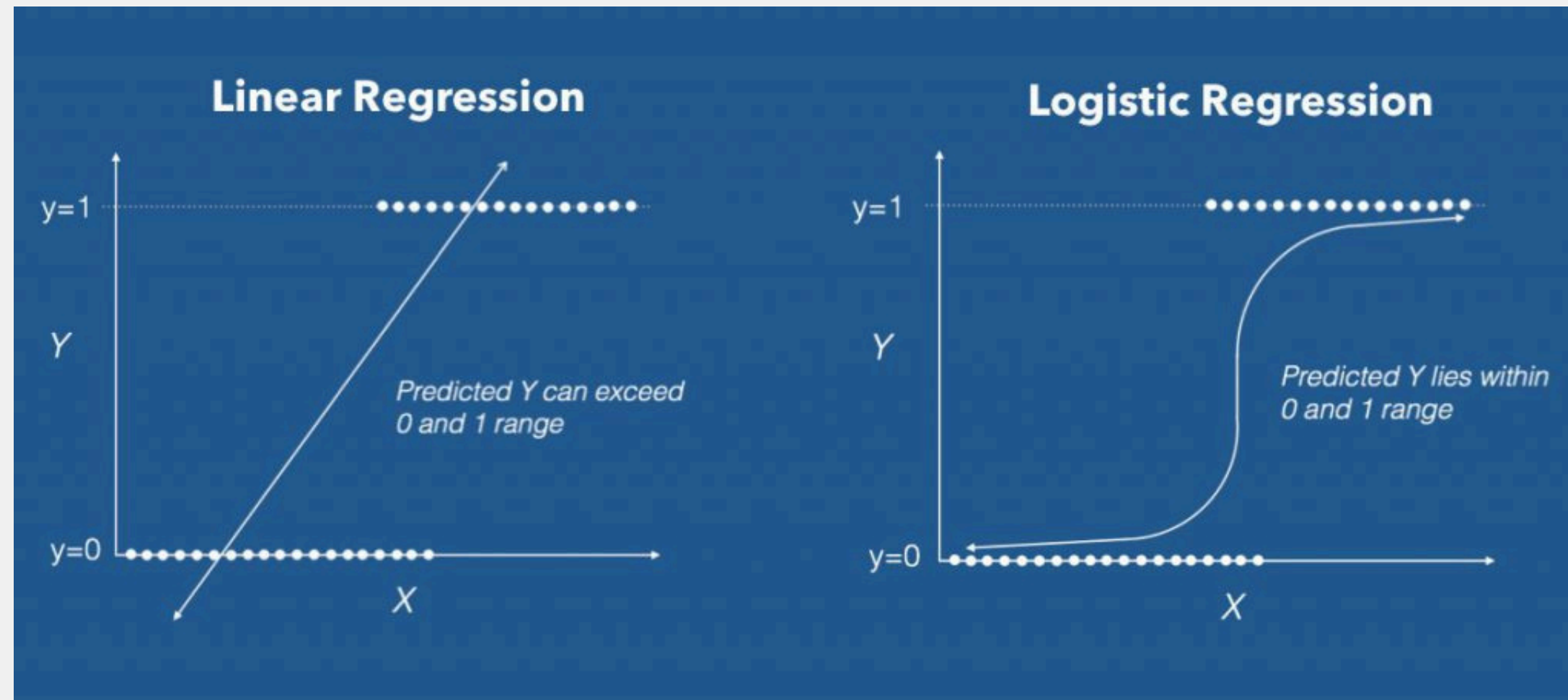
SELECTION FOCUS

- Focused on variables related to **customer retention** and **post-purchase behavior**
- Selected factors that help explain whether existing customers continue engaging with the platform or churn
- Prioritized behavioral, purchase-related, loyalty, and engagement signals

1. Customer profile: Customer segment, annual income, household size
2. Purchase behavior: Quantity, unit price, purchase frequency, days since last purchase
3. Loyalty & satisfaction: Loyalty score, loyalty tier, customer satisfaction
4. Engagement signals: Email open rate, social media engagement, newsletter subscribed
5. Risk indicators: Support tickets, return rate, payment delay days
6. Target variable: Churned

predicts a number **VS** probability between 0 and 1
-1000 , 2 Millions \$ **vs** Yes or no?

WHAT IS LOGISTIC REGRESSION



Step 1 — same linear combination:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Step 2 — wrap it in the sigmoid function:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

- **How likely this customer will churn?**
 - 0.5 and above = predict churn
 - below = predict stay
- Perfect for binary outcomes probability like yes/no, churn/stay, buy/don't buy

DATA FEATURE ENGINEERING

PREPARING THE DATA FOR LOGISTIC REGRESSION

FEATURE SELECTION

23 columns selected from full dataset

Dropped irrelevant identifiers & redundant variable to focus on behavioral and demographic signals predictive of churn

DATA QUALITY CHECK

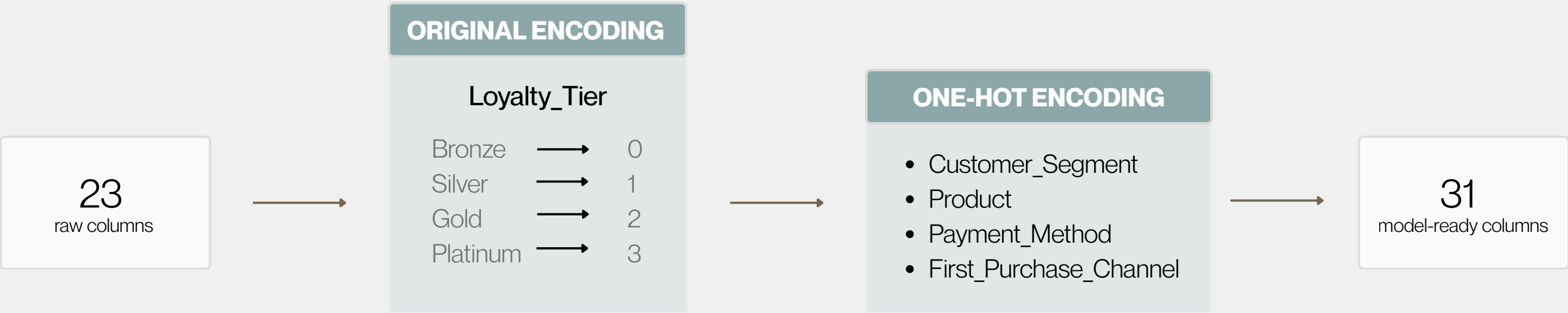
- Checked all 23 columns for nulls
- 0 missing values found
- No imputation required

Clean dataset - ready for encoding with no additional handling needed

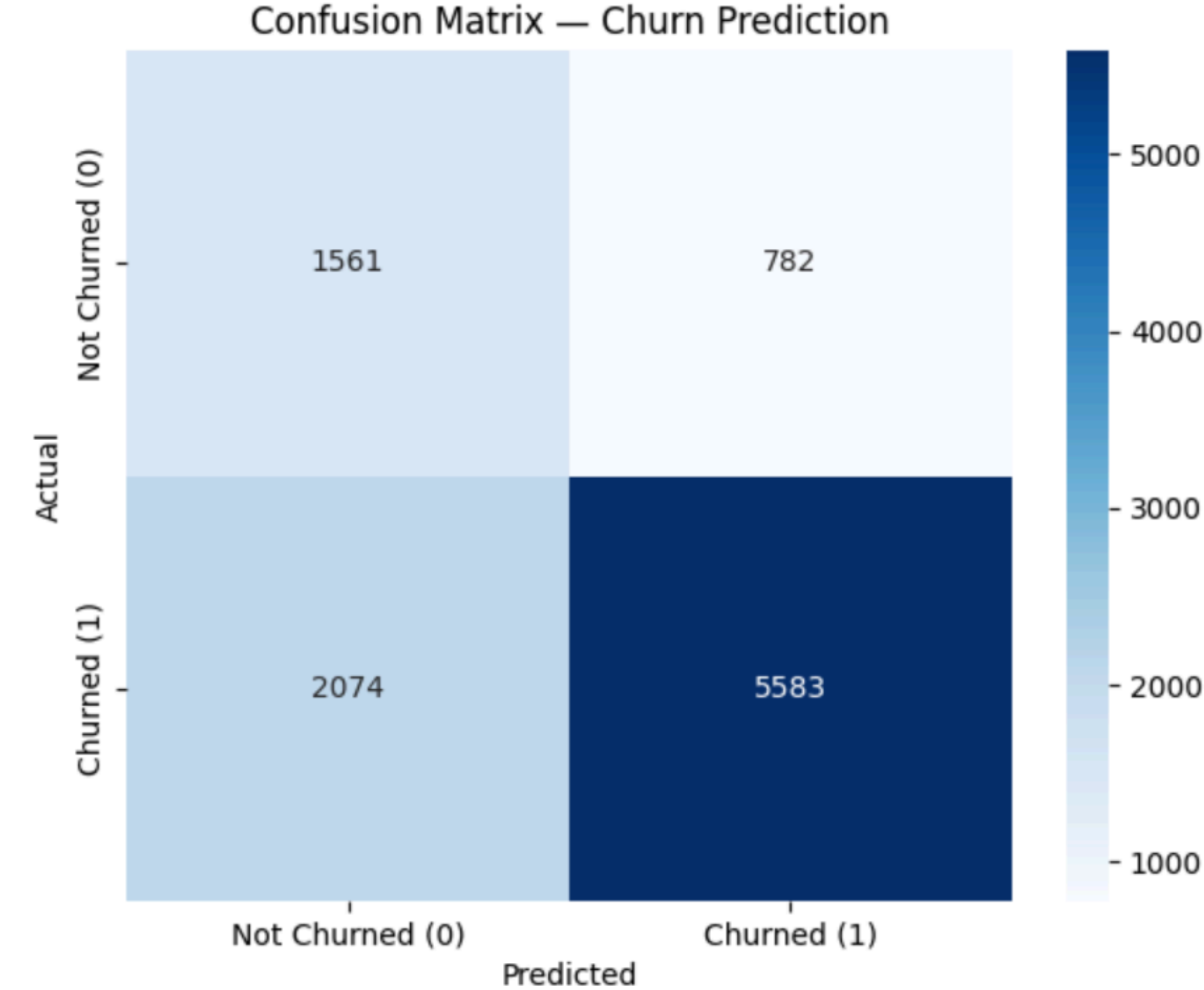
ENCODING RESULT

23 columns in → **31** columns out

+8 columns created from one-hot encoding 4 categorical variables



MODEL RESULTS



True Negatives (predicted stay, actually stay): 1561
False Positives (predicted churn, actually stay): 782
False Negatives (predicted stay, actually churn): 2074
True Positives (predicted churn, actually churn): 5583

TRAINED 80% :

TEST 20%

10 000 UNSEEN CUSTOMER

LIMITATIONS

Class imbalance (3:1 churn ratio) explains weaker non-churn performance

MODEL RESULTS

0.766

AUC-ROC

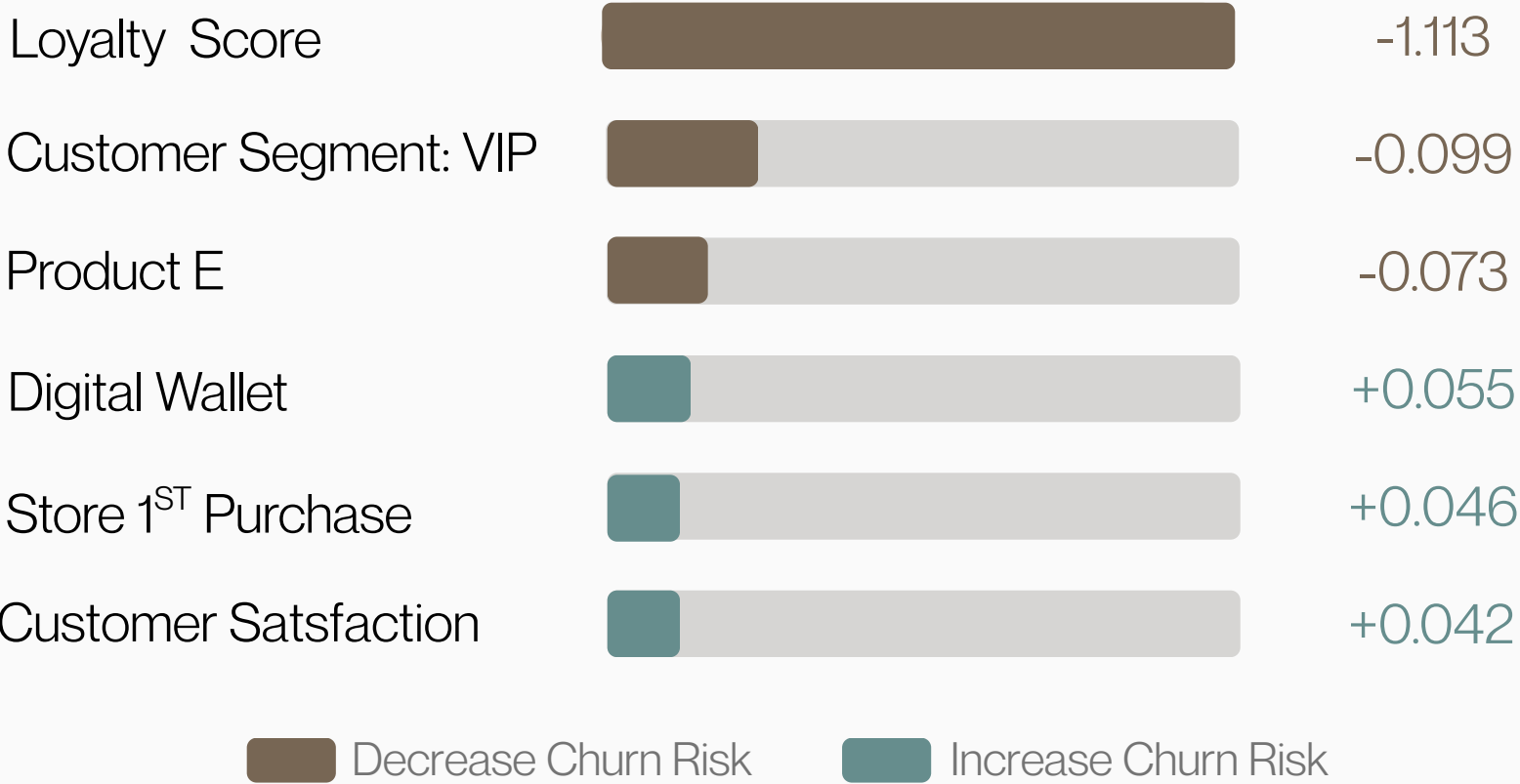
71%

ACCURACY

CLASS PERFORMANCE

CLASS	PRECISION	RECALL	F1-Score
Non Churner	43%	67%	52%
Churner	88%	73%	80%
Weighted Avg.	77%	71%	73%

TOP PREDICTORS



MODEL RESULTS

CLASS PERFORMANCE

Our model performs better at identifying churners than non-churners, reflecting a class imbalance in the data (churners outnumber non-churners 3 to 1)

Recall of 73% for churners means the model successfully catches 3 out of 4 actual churners

The weighted average F1 of 0.73 accounts for class size, giving an overall picture of model reliability across both customer groups

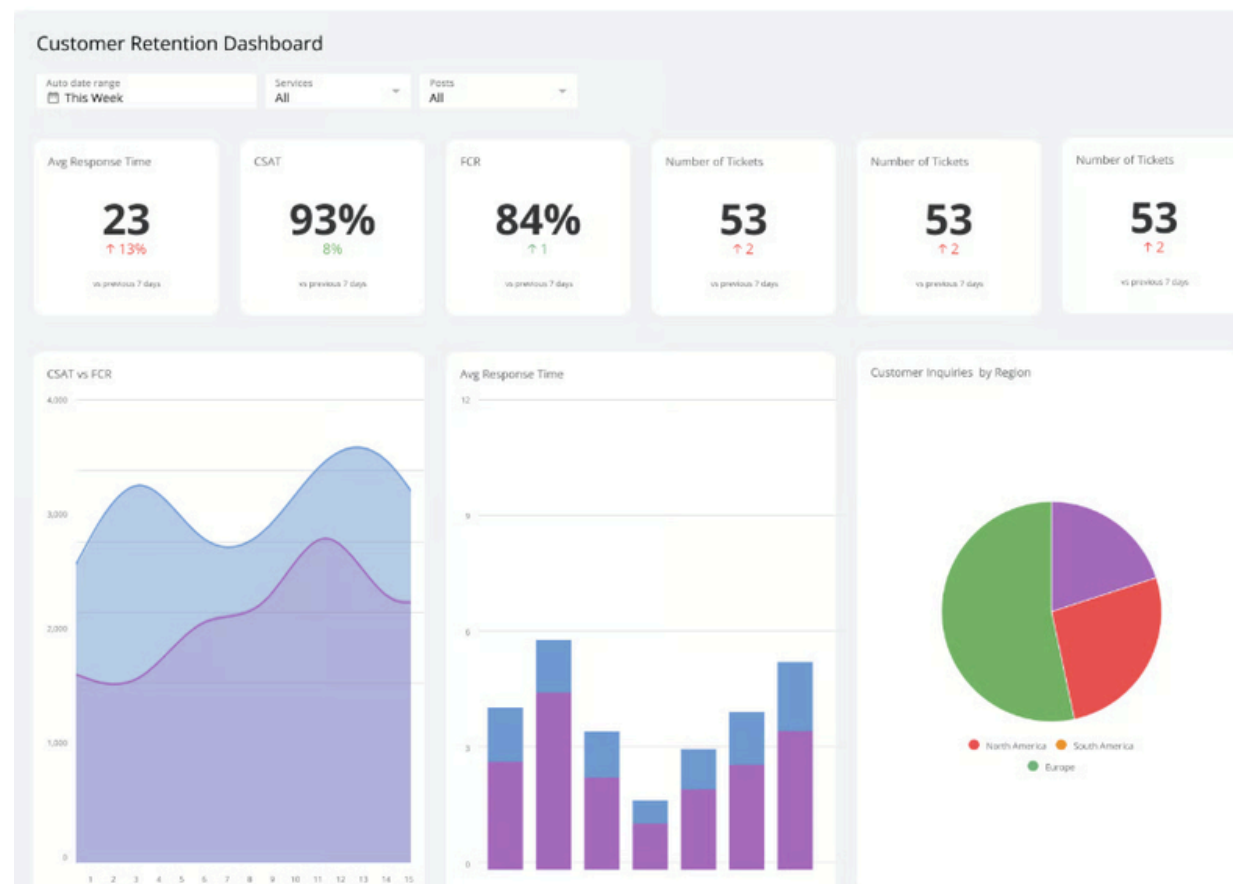
TOP FEATURES

Loyalty Score is the strongest predictor (customers with higher loyalty scores are significantly less likely to churn, making it the single most important variable in the model)

VIP customers and those who purchased Product E show negative coefficients, suggesting a naturally lower churn risk (potentially due to strong brand attachment or product satisfaction)

The large gap between Loyalty Score and all other features suggests the model is heavily driven by one variable

BUSINESS INSIGHTS



1

Prioritize loyalty programs

- Must invest in loyalty rewards,
- Tiered membership benefits
- Encourage repeat-purchase incentives

2

Segment-based intervention

Customer_Segment_VIP and Customer_Segment_Regular as influential variables.

VIP customers that has **rising Days_Since_Last_Purchase**

→ **prioritized** for personal outreach.

3

Deploy this model as a live scoring tool.

Automate run the model on CRM every week.

Dashboard: ranked list of at-risk customers

THANK YOU

Any questions?